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Evaluation of a Web-Based Simulation Tool for Self-Management Support in Type 1 Diabetes: A Pilot Study

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Abstract

Objective: To develop and evaluate a novel Web-based Simulation Tool (WST) that brings simulation technologies to people with Type 1 Diabetes (T1D), enabling unique patient-data interactions seamlessly on a daily basis.

Methods: A pilot clinical trial was conducted to assess system usability. The study consisted of one week of observation (Phase 1) and four weeks of interaction with WST (Phase 2). Responses to Technology Acceptance (TA) and Diabetes Distress Scale (DDS) questionnaires were collected, and follow-up interviews were conducted after Phase 2.

Results: Fifteen participants with T1D using Control-IQ® technology (age: 36±13 years, HbA1c: 6.5%±0.7%) completed all study procedures. Generated simulation models achieved a median Mean Absolute Relative Difference (MARD) of 6.8% [interquartile range, IQR: 5.1%,9.1%]. A decrease in expected benefits (likely explained by issues with the third-party data collection system) and an increase in expected burdens were observed. On a 1-5 scale, ease of use, trust, and usefulness scores were 3 [3,4], 4 [3,4], and 4 [3,4], respectively. Time below 70 mg/dL decreased between Phases 1 and 2 (1.6% [0.7%,3.7%] vs 0.8% [0.5%,3.0%]). A reduction in mean emotional burden was also observed (2.5±1.1 vs 2.1±0.8).

Conclusions: Results indicate that there was a learning curve to WST, but also that most participants trusted the system and found it useful in their diabetes care.

Significance: Simulation technologies like WST could be used by educators and patients to facilitate diabetes self-management, leading to better diabetes literacy and reducing associated distress.

Keywords

Diabetes distress; self-management; simulation; type 1 diabetes; web-based application

I. Introduction

BENEFITS of Intensive Insulin Treatment (IIT) on improving glucose control in people with Type 1 Diabetes (T1D), and as a consequence, on reducing the risk for developing diabetes complications, have been known since the Diabetes Control and Complications Trial (DCCT) [1]. For example, it has been reported that IIT reduces the risk of any cardiovascular disease event by 30% [2]. However, most people with T1D still do not achieve the glycated hemoglobin (HbA1c) target recommended by the American Diabetes Association (ADA) [3], in part due to the lack of education or self-management support [4]. Improving self-care practices is known to be fundamental to successful treatment, because the vast majority of the day-to-day care in diabetes is handled by the patients themselves and/or their families [5].

Advanced insulin therapy relies on key individual therapy profiles such as carbohydrate ratio (CR: grams of carbohydrate per unit of insulin), insulin sensitivity factor or correction factor (ISF or CF: mg/dL of glucose per unit of insulin), and insulin basal rate (IBR) [6]. Therapy profiles are not only used in standard open-loop basal-bolus treatments like Sensor-Augmented Pump (SAP) therapy, but also by automated insulin dosing systems like Control-IQ® technology [7]. Structured education programs are available to empower people with T1D to manage their disease and estimate individualized IBR, CR, and CF profiles [8]. However, despite these efforts, achieving tight glycemic control without increasing the risk for hypoglycemia still poses a difficult, life-long, optimization problem for people with T1D [9]. Periodic adjustments of IBR, CR, and CF profiles are needed based on review of Self-Monitoring Blood Glucose (SMBG) or Continuous Glucose Monitoring (CGM) data [8]. When a new pattern of glycemic risk is identified, patients need to tune their therapy profiles accordingly. This is a time-consuming, trial-and-error process that usually requires data to be downloaded from multiple devices, and experimentation that can take weeks to complete.

The rapid evolution of information technology has facilitated the management of chronic diseases [10], [11], including diabetes [12]. Among the vast number of technology-enabled applications [13], only few enable interactive simulations [14]-[20]. One example is the Karlsburg Diabetes Management System, KADIS® [17], [18]. This simulation-based decision support tool provides therapy recommendations based on an internal description of the patient's insulin-glucose dynamics. The core metabolic model is represented by a forth-order differential equation system that can be individualized using patient demographic

information, and a structured measurement plan that requires patients to make logbook entries of self-control data, such as time and amount of meal and insulin intakes. The process needed to identify risk patterns and simulate therapy adjustments is performed by operators of the Diabetes Service Center running the KADIS[®] program, who are specialized in the use of this tool. Acceptance rate of KADIS[®] as a decision support tool is 74%, and results from a one-year-long clinical trial showed a reduction in HbA1c of approximately 0.47% [17]. Another example of a simulation-based application is DiasNet [14]-[16]. Unlike KADIS[®], this system was also meant to be used by patients as an educational tool. Despite being developed in the early 2000s, DiasNet presented a novel concept where patients were able to simulate changes to their insulin doses and meal intakes by means of an Internet application equipped with a user interface. Glucose predictions were made using a two-compartment model of glucose metabolism implemented in a casual probabilistic network. Model parameters were adjusted using diabetes data that patients needed to manually enter into the system. The DiasNet application also showed effectiveness in reducing HbA1c when used to give insulin dose recommendations — a 1.2% reduction in HbA1c was reported in four studies [14].

Our goal is to further explore the impact of empowering self-management in T1D via proven simulation platforms and replay technologies. Given new technological advances, we were able to develop a novel Web-Based Simulation Tool (WST) that consolidates diabetes data and supports user driven alterations of historical meal and insulin records in a way that minimizes user effort. The foremost difference between KADIS[®] and WST is that patients have no interaction with the KADIS[®] system: Simulations are run sporadically by specialized operators who produce a report summarizing the *in silico* findings. On the contrary, WST interfaces directly with patients as our goal is for them to leverage the data they seamlessly generate. Consequently, patients are the ones who decide what and when to simulate in WST. The latter is based on the fact that people with diabetes have shown to better sustain the ongoing skills and behavioral changes necessary to manage their disease when they participate in their own care and receive self-management support [21], [22].

When the DiasNet application was used for patient empowerment, i.e., when the system was directly used by patients, difficulties were found with data entry that hindered the use of this pioneering system [15]. In this regard, an innovative feature of WST is that it is integrated with a hands-off data aggregation service — Tandem's t:connect[®] web application. The use of this technology allowed us to automate the process of uploading all device data to WST, without requiring users to manually perform this task on a regular basis. In this paper, we describe the design and deployment of WST, and present results from a pilot study where participants' acceptance of this environment was evaluated under normal conditions at home. The key hypothesis of this work is that by bringing simulation technologies to people with T1D through a robust, automatic, user-friendly platform, we can enable unique user-data interactions that can lead to a better understanding of their disease, and potentially, improved glucose control.

II. Materials and Methods

The general infrastructure of the system is depicted in Figure 1. Every day WST collects data from the users' t:slim X2™ insulin pumps automatically via Tandem's t:connect® web application. Field-collected data are first processed and then used to generate personalized models of the users' glucose metabolism — these models are needed to estimate the impact of meal and insulin changes on that day, i.e., to run the replay simulations. Both data collection and model optimization are defined as backend processes and do not require the users' intervention, who can interact with WST via a website portal. The website was designed to be fully responsive, i.e., the size and position of all the elements are adjusted automatically according to the size of the screen. Through it, users may request to see their data within a particular date range, manipulate their data — for instance, *what could have happened to my blood glucose if I had modified my basal insulin profile* — generate reports comparing their real and simulated data, and contact the study team via a contact form. Logs with records of system events and user interactions were automatically compressed, saved, and rotated on a daily basis for monitoring and posterior analysis. Technical details of how the system is deployed on Amazon Web Services (AWS) are provided in Appendix I.

A. Data flow

Every day a scheduler is called in the background to automatically coordinate the data flow from acquisition to storage.

1) Data fetching: Tandem's t:connect® mobile application uploads data from each insulin pump to a dedicated server every two hours or after a qualifying event. WST gets access to said server on a daily basis using the Tandem's t:connect® web application as follows:

- The system grabs an OAuth Access Token by making a POST request to Tandem's server, where the body of the request contains access credentials passed as x-www-form-urlencoded data. Credentials are stored as environment variables on WST's server, and the Access Token expires after 8 hours.
- The system grabs a zip file for each active user via a GET request to Tandem's server using the issued Access Token as a "Bearer Token" and the pump model and serial numbers, which are protected through symmetric encryption.
- Received files are saved to memory and then decompressed to read the de-identified JavaScript Object Notation (JSON) data.

2) Data processing: JSON files contain glucose, insulin, meal, and therapy profile (IBR, CR, and CF) records that WST reads and processes as follows: (i) samples are sorted by ascending time; (ii) timestamps are synchronized; (iii) duplicate records are removed; (iv) gaps are located and their extension measured; (v) qualifying events are detected (extended boluses, declined corrections, temporary basal rates, and overwritten doses); (vi) gaps are filled depending on the type of data (for instance, linear interpolation for CGM data, and previous values for IBR); and (vii) days of data are classified as either playable (a model can be obtained) or non-playable according to a Data Quality Assessment (DQA) checklist. The

DQA includes glucose and insulin gaps no longer than 2 hours during the day and 3 hours during the night (12:00 AM to 06:00 AM), and the existence of IBR, CR, and CF profiles. In addition, WST checks whether informed insulin doses were then aborted to update the insulin records with the values that were actually injected. In case the user suspends prandial doses (doses associated with meals), WST analyzes the glucose rate of change around the informed mealtime. If a negligible deviation is detected, the meal record is removed.

3) Meal detection algorithm: In real-life conditions, meals are not usually recorded in a timely manner. There are cases where a meal is reported before or after the actual mealtime, or not reported at all. From a modeling perspective, misalignments between glucose deviations and meal entries could be erroneously explained by design variables during the optimization procedure. For instance, a late meal announcement, i.e., when the meal is recorded after the actual mealtime, could be interpreted as a change in insulin sensitivity between glucose started to increase and the meal is informed to the model. For this reason, after cleaning the data, a meal detection algorithm is run to adjust the time of announced meals, and detect potential unannounced meals and hypo-treatments. A detailed pseudo-code is presented in Appendix II, but its main functions are summarized here. Given a one-day-long data set, the algorithm: (i) captures the mealtimes of reported meals; (ii) estimates the first and second derivatives of the glucose signal; (iii) saturates them at zero and multiplies them together (μ); (iv) adjusts the mealtime of each reported meal by looking for a peak in μ around the informed time and moving the mealtime to the base of the potential peak; (v) finds all the peaks in μ that meet certain criteria with respect to height ($P_m = 0.0075 \text{ mg}^2/\text{dL}^2/\text{min}^3$), prominence ($P_p = 0.0025 \text{ mg}^2/\text{dL}^2/\text{min}^3$), and separation (30 min); (vi) defines potential mealtimes at the peaks' bases (fixed size based on historical records); (vii) classifies an event as a hypo-treatment based on a glucose threshold and the size of the peak; (viii) if the event is not recognized as a hypo-treatment, then the algorithm evaluates the glucose deviation to determine if it can be classified as a meal; and (ix) appends the vector of detected, unannounced meals to the vector of adjusted, informed meals, eliminating repetitions. The algorithm was designed and tuned using data sets that consisted of a mix between real data from previous supervised hotel admissions ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT02558491) identifier: [NCT02558491](https://clinicaltrials.gov/ct2/show/study/NCT02558491)) and *in silico* data generated with the UVA/Padova simulator [23], where different conditions including early, perfect, late and no meal announcements were recreated under fully-controlled conditions. A grid-search procedure for parameter tuning (for instance, P_m and P_p) was conducted and the combination that maximized the algorithm's sensitivity and precision was selected as optimal.

4) Virtual image generation: WST relies on the reconstruction-replay framework that was presented by the authors in [24]. In its core, there is a model of glucose metabolism that consists of a time-invariant version of the Food and Drug Administration (FDA)-accepted UVA/Padova model combined with an auxiliary input ω that captures daily variations in insulin sensitivity. The auxiliary input modulates the impact of insulin on endogenous glucose production and insulin-dependent glucose utilization, and is represented as a truncated Fourier series:

$$\omega(t) = \phi_1 + \sum_{i=1}^h [\phi_{2i} \cos(h\omega_0 t) + \phi_{2i+1} \sin(h\omega_0 t)] \quad (1)$$

with a fundamental period of 1 day (ω_0), h harmonics, and a design variable vector $\phi = (\phi_1, \dots, \phi_{2h+1})^T$. To address overfitting, exhaustive parameter identifiability analysis that included global ranking of parameters, collinearity index, and Akaike's final prediction error was performed offline to determine the set of most-impacting, low-correlated model parameters θ that combined with ϕ led to the most parsimonious solution.

For each day of data that met the DQA requirements, i.e., for each playable day, the parameter vector $\kappa = \begin{pmatrix} \zeta \\ \phi \end{pmatrix} = \begin{pmatrix} \ln(\theta) \\ \phi \end{pmatrix}$ was personalized by means of the following Bayesian-based optimization procedure:

$$\kappa^* := \arg \min_{\kappa} f(\kappa) \quad (2)$$

with cost function:

$$f(\kappa) = (\mathbf{y} - \mathbf{y}_p)^T \Sigma_v^{-1} (\mathbf{y} - \mathbf{y}_p) + (\zeta - \mu_\zeta)^T \Sigma_\zeta^{-1} (\zeta - \mu_\zeta) \dots + (\phi - \mu_\phi)^T \Sigma_\phi^{-1} (\phi - \mu_\phi) \quad (3)$$

where $\mathbf{y} - \mathbf{y}_p$ is difference between glucose measurements and predictions, Σ_v represents the covariance matrix of measurement errors, μ_ζ and Σ_ζ are the model parameter mean and covariance matrix in logarithmic form, and μ_ϕ and Σ_ϕ are used to penalize deviations from $\omega(t) = 1$, the nominal case. Thus, the cost function takes into account model fitting, distance to the simulator's distribution, and regularization of the auxiliary input. Moreover, lower and upper bounds for the design variables were added to the optimization problem formulation to reduce the search space. A complete description of this methodology can be found in authors' previous work [24]. The optimization problem was implemented in MATLAB 2018b (MathWorks, Natick, MA) assuming multiple initial conditions, and parallelized on WST's high-performance computing (HPC) service.

B. System functionality

The first page that users look at is the Login page, where they can gain access to WST by entering their e-mail address and password. Once logged in, users are redirected to a Dashboard page that is the main screen of the application, where they can visualize their data and run simulations (see Figure 2). WST is also equipped with (i) a Support page where users can review frequently asked questions, watch tutorial videos, and run an interactive example that guides them through learning the system; (ii) a Contact page where users can request more information about system functionalities; and (iii) a Profile page where users can find basic information about their accounts. As shown in Figure 2, the Dashboard page is divided into a Display panel and a Replay panel.

1) Display panel: In the Display panel, users can select a particular date range using a calendar. They can study a single day or combine multiple days for more systematic reviews.

They can also control the amount of information displayed on the screen, showing or hiding extra indices like glucose variability, and risks for hypo- and hyperglycemia.

2) Replay panel: In the Replay panel, users can select various pump's parameters, like IBR, CR, and CF profiles, a particular time interval, and modulate the signal amplitude by just moving a slider. A similar procedure is proposed to interact with meals, where users can alter the time and size of all their meals at the same time, or on a one-by-one basis. To perform a simulation, users just need to tap the *Run* button. This will make the system carry out a replay simulation, which status is monitored by a progress bar. Results of the replay simulation are shown along with the reconstructed data to facilitate the comparison between these two. Once the simulation is done, users can generate a pdf report highlighting all made changes, and save their new insulin/meal configuration — WST can store replay configurations temporarily, allowing users to easily compare up to three different replays.

C. Study design

WST was evaluated in a single-arm, uncontrolled, pilot clinical trial of adult subjects at home ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT04439903) identifier: [NCT04439903](https://clinicaltrials.gov/ct2/show/study/NCT04439903)). The research protocol and data security plan was approved by the University of Virginia Institutional Review Board (IRB-HSR 200157). The primary aim of the study was to assess WST's usability after one month of system use. Main eligibility criteria included T1D for at least one year, current user of the Tandem t:slim X2™ insulin pump, and willingness to interact with a computer program. Major exclusion criteria included inability to read and complete questionnaires or interact with a computer program, history of a seizure disorder (except hypoglycemia seizure), and pregnancy. The reason for requiring Tandem users was to leverage the Tandem's t:connect® web application to consolidate, aggregate, and transmit data automatically from the participants' insulin pumps to our system's database. Otherwise, participants would have needed to upload their data to WST manually on a daily basis. This could have affected their attitude towards using the system as most people with T1D using devices download their data very infrequently [25].

The concept of the technology was first discussed with subjects when they went through the informed consent process, at which time the language discussing the technology is decidedly neutral. Once enrolled, participants attended a 30-60 minute training session that consisted of an IRB approved slide presentation, describing all the different features and functions of the UI, and access to a simulated subject's data for a hands-on demonstration of the insulin and meal controls on the WST website. At the end of the session, the subjects were provided with the slide deck as well as an instruction manual from which the slides were devised. The first week after training (Phase 1) was purely observational, and data collected during that time was used to estimate the baseline glucose metrics. During the following four weeks (Phase 2), participants were asked to interact with WST at least once a week and were informed to speak with their Health Care Providers (HCPs) prior to making changes to their treatment parameters. Responses to questionnaires were collected pre and post system use. In addition, a semi-structured, open-end, follow-up interview with a study team member was conducted about two weeks after completing Phase 2 to allow the participants to describe their experience with WST. The interview was conducted virtually and consisted of seven

questions exploring how the experience with the system was, what should be improved, and what they felt was missing and should be implemented or improved.

The primary outcome was a specifically curated technology acceptance (TA) questionnaire based on [26]. Items in this questionnaire were scored on a 5-point Likert scale, producing two scores: Benefits and Burdens. Each score was computed as the mean item score converted into a percentage.

The secondary aim was to track changes in treatment satisfaction as measured by the Diabetes Distress Scale (DDS), a 17-item scale that yields a total diabetes distress score and 4 subscale scores: Emotional burden, regimen distress, interpersonal distress, and physician distress [27]. Mean item scores <2.0 , between 2.0 and 2.9, and >3.0 , are considered little to no distress, moderate distress, and high distress, respectively. A sample of the interview guide and questionnaires can be found as Supplementary Material. Finally we explored potential impact on glucose control as characterized by CGM-based outcomes.

As a pilot study, this clinical trial was not formally powered to assess effect, but p-values (significance level at 0.05) will be reported for a more comprehensive analysis of the results. Paired t-test will be used for normally distributed data and Wilcoxon signed-rank test, otherwise. Descriptive statistics will be also considered to infer size and direction of treatment effect. The mean and standard deviation (SD) are reported when the distributions of data are normal, and the median and interquartile range [IQR], otherwise.

III. Results

Fifteen adult participants (four men and eleven women) using Control-IQ[®] technology completed all study procedures (18 enrollments with 3 screen failures). Mean demographics of the study group were as follows: age, 36 ± 13 years; HbA1c, $6.5\% \pm 0.7\%$; weight, 71.2 ± 18.5 kg; and total daily insulin, 35.4 ± 11.8 U.

A. Data processing and system use

The identification procedure was executed only for playable days, i.e., days that met the DQA criteria defined in Subsection II-A.2. In other words, days with large data gaps or missing insulin profiles were dropped from this analysis, considering there were not enough data to obtain a reliable model for replaying purposes. In this context, WST was able to generate models from 86.4% of available days of data, achieving a median root mean square error (RMSE) of 14.1 mg/dL [9.6 mg/dL, 18.3 mg/dL], and median mean absolute relative difference (MARD) of 6.8% [5.1%, 9.1%] when compared with the real glucose measurements. Using the error grid analysis (EGA) as another performance instrument [28], glucose estimates are plotted against CGM measurements on a grid divided into five zones that represent different levels of accuracy: Accurate (A-zone), benign deviation (B-zone), overestimation of hypo/hyperglycemia (upper/lower C-zone), failure to detect hypo/hyperglycemia (lower/upper D-zone), and erroneous estimation (E-zone). Results from this analysis indicate that the median percentage of reconstructed glucose values that fell into the clinically acceptable A- and B-zones was 99.7% — A-zone, 95.5% [88.5%, 99.0%]; B-zone,

4.2% [1.0%,10.4%]; C-zone, 0.0% [0.0%,0.0%]; D-zone, 0.0% [0.0%,0.4%]; E-zone, 0.0% [0.0%,0.0%]. Average weighted residuals are illustrated in Figure 3.

Mean spent time using WST per participant was 63.1 min (42.6 min), or approximately 15 min, per participant per week. In terms of interactions with the Display and Replay panels, median numbers of click events on the calendar and *Run* button were 15 [11,28], and 29 [17,42], respectively. Users preferred to access the app via Desktop computers (85.7% of visits to WST), while 14.3% of visits came from mobile devices.

B. Technology Acceptance

TA analysis for the overall group of participants indicates a decrease in expected benefits ($61.9\% \pm 13.3\%$ vs $52.0\% \pm 23.5\%$, $p=0.045$) and increase in expected burdens ($27.5\% \pm 16.3\%$ vs $39.7\% \pm 17.6\%$, $p=0.030$). To understand this result is important to highlight that the Tandem's t:connect[®] web application was down for one month due to maintenance work. This issue considerably affected four participants who did not have the opportunity to interact with up-to-date data during a significant part of Phase 2. If we isolate this problem by removing the affected participants from the analysis, expected benefits remained constant between beginning and end of the study ($65.3\% \pm 12.5\%$ to $63.1\% \pm 14.4\%$, $p=0.566$), however, a non-significant increase in the burden score is still observed: from $25.0\% \pm 13.7\%$ before system use to $33.9\% \pm 16.0\%$ afterwards ($p=0.180$). For example, the mean score of *Feeling less burdened in managing diabetes while using WST* was 50.0% pre- and post-system use in the overall group, but 50.0% vs 61.4% ($p=0.359$) in the unaffected subgroup. On the other hand, the mean score of *WST is more complicated than my usual CGM report* was 40.0% vs 58.3% ($p=0.022$) in the overall group, and 38.6% vs 52.3% ($p=0.140$) in the unaffected subgroup. More details on this subject will be covered in the analysis of the follow-up interviews. In terms of how robust WST is, participants did not report “glitches” and “bugs”, and the associated mean score was low pre- and post-system use (31.7% vs 26.7%, $p=0.425$).

Similar results can be observed with respect to participants' evaluations regarding ease of use, trustworthiness, and usefulness of WST (see Table I).

C. Diabetes distress scale

Analysis of the DDS shows that although participants presented little or no distress at enrollment, a slight reduction in the median total score was observed: (1.9 [1.5,2.0] vs 1.8 [1.3,2.4], $p=0.306$). This reduction was particularly driven by a reduction in emotional burden (2.5 ± 1.1 vs 2.1 ± 0.8 , $p=0.492$) and regimen distress (1.6 [1.4,2.5] vs 1.4 [1.2,2.3], $p=0.369$), which are two subscales of the DDS that could be linked to the use of WST. Note that emotional burden is associated with feeling overwhelmed, frightened, or fearful about managing the demands of diabetes over time, and regimen distress, with feeling you are failing by not managing your diabetes properly. For instance, the mean score of *Feeling overwhelmed by the demands of living with diabetes* changed from 2.6 to 2.0 ($p=0.211$), and the mean score of *Not feeling confident in my day-to-day ability to manage diabetes*, from 1.9 to 1.7 ($p=0.462$).

D. Glucose control

Overall percentage of time in the range [70,180] mg/dL was virtually the same in Phase 1, 79.5%±13.0%, and Phase 2, 79.2%±11.3% ($p=0.882$) — one participant was excluded from this analysis due to testing COVID positive during Phase 2 and reporting abnormally high blood glucose values. A slight increase in overall time in the range [70,250] mg/dL was detected (94.3% [91.3%,95.7%] vs 96.2% [92.1%,97.9%], $p=0.447$), particularly fueled by a reduction in time below 70 mg/dL (1.6% [0.7%,3.7%] vs 0.8% [0.5%,3.0%], $p=0.517$). These changes can be observed especially overnight (time in 70-250 mg/dL: 92.8% [88.3%,98.2%] vs 97.2% [91.8%,99.6%], $p=0.233$; time <70 mg/dL: 1.7% [0.0%,4.2%] vs 0.6% [0.0%,2.5%], $p=0.127$), and for those participants in this analysis (4) who reported modifying their pumps' settings based on WST simulations (time in 70-250 mg/dL: 90.5%±2.7% vs 93.1% ±6.7%, $p=0.503$; time <70 mg/dL: 1.7% [0.0%,6.5%] vs 0.0% [0.0%,0.9%], $p=0.253$). This group of four participants (i-iv) reported changing: (i) IBR and CF once, (ii) IBR once, (iii) IBR twice and CR once, and (iv) IBR twiced and CF once. Main glycemic outcomes are summarized in Table II, and mean time-in-target results are depicted in Figure 4.

Same results can be observed from the risks for hypo- and hyperglycemia. Overall Low Blood Glucose Index (LBGI) and High Blood Glucose Index (HBGI) [9] in Phases 1 and 2 were 0.6 [0.3,0.9] vs 0.5 [0.2,0.9] ($p=0.730$) and 4.2±2.6 vs 4.3±2.2 ($p=0.883$), respectively. Overnight indices were 0.6 [0.1,1.0] vs 0.4 [0.2,1.0] ($p=0.427$), and 3.6 [1.4,6.1] vs 2.5 [1.1, 6.8] ($p=0.769$), and particularly for the participants who modified their pumps' settings based on WST simulations were 0.5 [0.0,1.5] vs 0.1 [0.0,0.3] ($p=0.225$), and 6.1±4.5 vs 6.7±4.5 ($p=0.538$). An illustrative example from a participant who reported making therapy changes based on WST information is presented in Figure 5. As shown, lowering the IBR profile before midnight helped the participant reduce the risk for hypoglycemia overnight.

E. Follow-up interviews

Participants' responses to the main questions of the follow-up interview are summarized in Figure 6. When participants were asked whether they enjoyed using WST, most of them answered yes (10), 3 moderately, and 2 no. Examples of comments from those who answered yes are: *"it helped me figure out adjustments to have a more stable blood sugar rather than peaks and valleys"*, and *"it was a different kind of educational tool, one that I hadn't experienced before. It added something new to patient care, for sure."* Among the participants who answered moderately, one reported already having a finely tuned pump and not much to explore, one desired there were more features focused on physical exercise, and the last one reported not needing this technology, but seeing WST as a great benefit, especially for those people with greater glycemic variability. Among the participants who answered that they did not enjoy using WST, one said that the main problem was the need to carry a study phone with them — there were network data usage issues between the participant's personal smartphone and the t:connect mobile application — and the other participant reported being more accustomed to other applications.

In agreement with their responses to the TA questionnaire, a few participants reported needing some time to get comfortable with WST's settings, but the majority of them (9)

considered that WST was easy to learn (Q2: “*Was it easy to learn how to use WST?*”). This was somehow supported by participants’ responses to question Q3: “*Do you think you have received enough training before the beginning of the trial?*”, where 14 of them considered that the training they received was sufficient and effective.

At the end of the interview when participants were asked whether they would consider using WST regularly as part of their diabetes care, 10 answered yes, 1 said that probably would not, but saw the potential of it for other people who struggle more with their diabetes, and 4 answered that probably not, unless it was equipped with other features like automatic therapy recommendations. It is worth mentioning that the addition of automatic therapy recommendations is being studied in an ongoing clinical trial (project number: 2R01DK085623-10, [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT04784637) identifier: [NCT04784637](https://clinicaltrials.gov/ct2/show/study/NCT04784637)).

IV. Discussion

Empowering effective self-management of T1D is associated with better glycemic control and quality of life [8]. The objective of this work was to facilitate this process by bringing state-of-the-art software and simulation technologies directly to people with T1D. To this end, we developed and tested the novel web-based, educational platform, a.k.a. WST, which allows people with T1D to fully leverage the data they generate. Through WST, they cannot only visualize their diabetes data, but also quickly and easily explore changes to their insulin therapy parameters and meals.

Although similar approaches have been implemented by previous systems like KADIS[®] [17], [18] and DiasNet [14]-[16], to the best of the authors’ knowledge, WST is the first ever cloud-based system capable of bringing simulation support to people with T1D on a daily basis. Results from the pilot study indicate that WST can guarantee continuity in model personalization (86.4% success rate) with clinically acceptable fitting (median MARD of 6.8%). Another important aspect of WST is its flexibility and autonomy from the user standpoint. Its responsive design and backend workflow offer users a seamless experience on any device. Manual uploading of data could be a major factor in users’ discontent with this type of applications. This burden on users was removed by means of a completely hands-off data collection and processing procedure that leverages the Tandem’s t:connect[®] web application.

Although participants reported there is a learning curve to WST (ease of use: 3 [3,4]), which was somehow anticipated given the novel nature of this tool, they also claimed that the training was adequate (14 out of 15), and the overall message was “*the more you learn WST the better your dial things in*”. Most participants found WST useful (usefulness: 4 [3,4]), and would consider using it regularly as part of their diabetes care. It is also worth highlighting that a slight reduction in diabetes distress was found after system use (emotional burden: 2.5 ± 1.1 vs 2.1 ± 0.8 and regimen distress: 1.6 [1.4,2.5] vs 1.4 [1.2,2.3]) accompanied by a slight improvement in glucose control, particularly overnight, and especially for those participants who reported modifying their pumps’ settings based on WST information. No significant correlation was observed between interaction time and main outcomes. Little negative correlation (magnitude of correlation coefficient <0.3) was detected with initial

emotional burden. Within the unaffected subgroup, low negative correlation (magnitude of correlation coefficient <0.5) was found between interaction time and perceived benefits, meaning that longer interactions could, in fact, indicate potential difficulties in interpreting/using the application.

Paradoxically, the hands-off data collection method of WST could also be pointed out as a limitation, since it limits the system use to people on Tandem t:slim X2™ insulin pumps, and makes it susceptible to external issues that can affect its usability, as demonstrated in the Results. It should also be added that all participants in our study were Control-IQ® users with moderately low diabetes distress and tight glucose control (HbA1c: $6.5\% \pm 0.7\%$), two facts that along with the narrow age-range could have limited the generalizability of the findings. Although the trend of obtained results pointed in the right direction, caution should be exercised with this analysis given the absence of a control group, the pilot nature of the study, and its main purpose, which was to evaluate participants' interest and acceptance of WST. It should be stressed that diabetes self-management involves an intricate interplay between many factors that are not addressed by this simulation tool. WST approaches this problem from one angle that is to provide educational support through enabling user-data interactions. Moreover, although the concept of WST can be replicated in other types of diabetes, several aspects should be redesigned to account for the specific characteristics and clinical requirements of the target population.

V. Conclusion

In this article, we presented a novel web-based simulation tool, WST, that allows people with T1D to interrogate and manipulate their data seamlessly on a daily basis. Results from a pilot study with fifteen adults with T1D indicated that there is a learning curve to the system, but that most participants trusted it and found it useful. Slight reduction in diabetes distress and improvement in glucose control were also observed, suggesting that simulation technologies could be beneficial for people with T1D by empowering them to manage their own disease. Future controlled studies should be conducted to validate the use of this technology.

Acknowledgments

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Appendix I: System deployment

WST's server is deployed on an Amazon EC2 r5.xlarge instance (vCPU:4, memory: 32 GiB, platform: Ubuntu). The core system is a Flask application that generates the dynamic content, including simulation data and glycemic metrics. Nginx is the front facing web server that handles the Hypertext Transfer Protocol (HTTP) requests that come in from the Internet, providing the static and dynamic content back to the user. Between Nginx and Flask lives Gunicorn, which is a Web Server Gateway Interface (WSGI) HTTP server that receives requests from Nginx and routes them to the Flask application. To handle multiple concurrent HTTP requests, i.e., to support multiple concurrent users, Gunicorn

was configured to use a mix of workers and threads. WST's database is on an Amazon db.t2.medium instance class (vCPU: 2, memory: 4 GiB, engine: MySQL), and interacts with the Flask application through SQLAlchemy. Access to WST's database is handled by a Virtual Private Cloud (VPC) security group that controls both inbound and outbound traffic. The website has also been secured by a Secure Sockets Layer (SSL) certificate installed on the Nginx server to encrypt data transferred between users and WST. WST's High-Performance Computing service is supported by a Matlab cluster that comprises a head node (the job scheduler) and compute nodes (workers). The number of compute nodes is scaled up or down automatically to balance computing power and associated cost based on demand.

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Appendix II: Meal detection algorithm: Pseudo-code

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Input: Arrays of daily CGM values, $\mathbf{g} = (g_1, \dots, g_n)$, and reported meals, $\mathbf{m} = (m_1, \dots, m_n)$, sampled every 5 min with $n = 288$.

Output: Updated array of meals.

- 1: Capture the indices of reported meals $\mathbf{I}_m = \{k \in \mathbb{N}_1^{288} : m_k > 0\}$.
- 2: Estimate the element-wise first and second derivatives of $\mathbf{g}$ ($\hat{\mathbf{g}}$ and $\hat{\mathbf{g}}'$) by means of a Savitzky-Golay filter of polynomial order 3 and frame length 13.
- 3: Saturate $\hat{\mathbf{g}}$ and $\hat{\mathbf{g}}'$ at zero.
- 4: Calculate $\mu = \text{sat}(\hat{\mathbf{g}}) \odot \text{sat}(\hat{\mathbf{g}}')$ with $\odot$ denoting the element-wise product operation.
- 5: **for** each element I_{m_i} of $\mathbf{I}_m$ $\diamond$ Adjust the time of reported meals
- 6: Find a potential peak index in μ around I_{m_i} , $I_{\mu_i} = \{k \in S_1 : \mu_k \leq \mu_{I_{m_i}}, \forall k \in S_1\}$ with $S_1 = [I_{m_i} - i_{1,min}, I_{m_i} + i_{1,max}]$ and $i_{1,min}$ and $i_{1,max}$ being the lower and upper bound indices of the search area.
- 7: Move I_{m_i} closer to the peak's base, $I_{\mu_i} = \{k \in S_2 : \mu_{k+1} - \mu_k \leq \mu_{s+1} - \mu_s, \forall s \in S_2\}$ with $S_2 = [I_{\mu_i} - i_{2,min}, I_{\mu_i}]$ and $i_{2,min}$ being the lower bound index of the search area.
- 8: **end for**
- 9: Identify potential unannounced meals at peak indices $\mathbf{I}_d$ in μ with height, prominence, and separation greater than $P_m = 0.0075 \text{ mg}^2 / \text{dL}^2 / \text{min}^3$, $P_p = 0.0025 \text{ mg}^2 / \text{dL}^2 / \text{min}^3$ and 30 min, respectively.
- 10: Evaluate μ at $\mathbf{I}_d$ to extract peak heights $\mathbf{P}_h$.
- 11: **for** each element I_{d_i} of $\mathbf{I}_d$ $\blacktriangleright$ Append detected unannounced meals / hypo-treatments
- 12: Move I_{d_i} closer to the peak's base, $I_{d_i} = \{k \in S_3 : \mu_{k+1} - \mu_k \leq \mu_{s+1} - \mu_s, \forall s \in S_3\}$ with $S_3 = [I_{d_i} - i_{2,min}, I_{d_i}]$.
- 13: Compute glucose deviations between peak's base to 60 minutes ahead $\Delta g = g_{I_{d_i}+12} - g_{I_{d_i}}$.
- 14: **if** $g_{I_{d_i}} \leq g_t$ **and** $P_{h_i} \geq P_{ht}$ **then** $\blacktriangleright$ g_t and P_{ht} are glucose (90) and peak (0.01) thresholds, respectively
- 15: **if** I_{d_i} is more than 3 samples apart from other detected hypo-treatments **then**
- 16: Append 20 grams and I_{d_i} to $\mathbf{m}$ and $\mathbf{I}_m$, respectively.
- 17: **end if**
- 18: **else if** $\Delta g \geq \Delta g_t$ **then** $\blacktriangleright$ Δg_t is the glucose deviation threshold (40)
- 19: **if** I_{d_i} is more than 9 samples apart from other meals **then**
- 20: Append m_s and I_{d_i} to $\mathbf{m}$ and $\mathbf{I}_m$, respectively. $\blacktriangleright$ m_s is the average meal size obtained from historical records or 0.7g / kg if there are not
- 21: **end if**
- 22: **end if**
- 23: **end for**
- 24: Sort $\mathbf{m}$ and $\mathbf{I}_m$ by $\mathbf{I}_m$ in ascending order.
- 25: **return** $\mathbf{m}$ and $\mathbf{I}_m$

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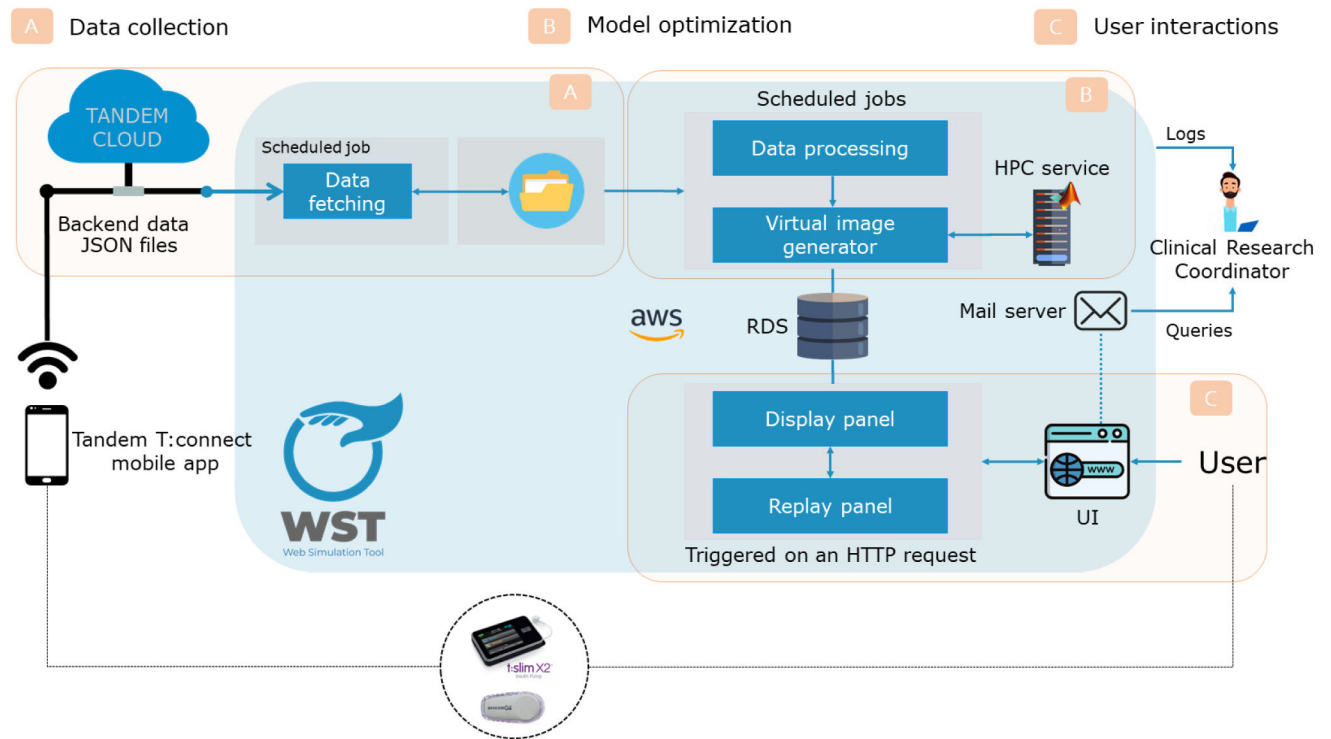


Fig. 1. General infrastructure of WST. JSON: JavaScript Object Notation, HPC: High-Performance Computing, AWS: Amazon Web Services, RDS: Relational Database Service, UI: User Interface, and HTTP: Hypertext Transfer Protocol.

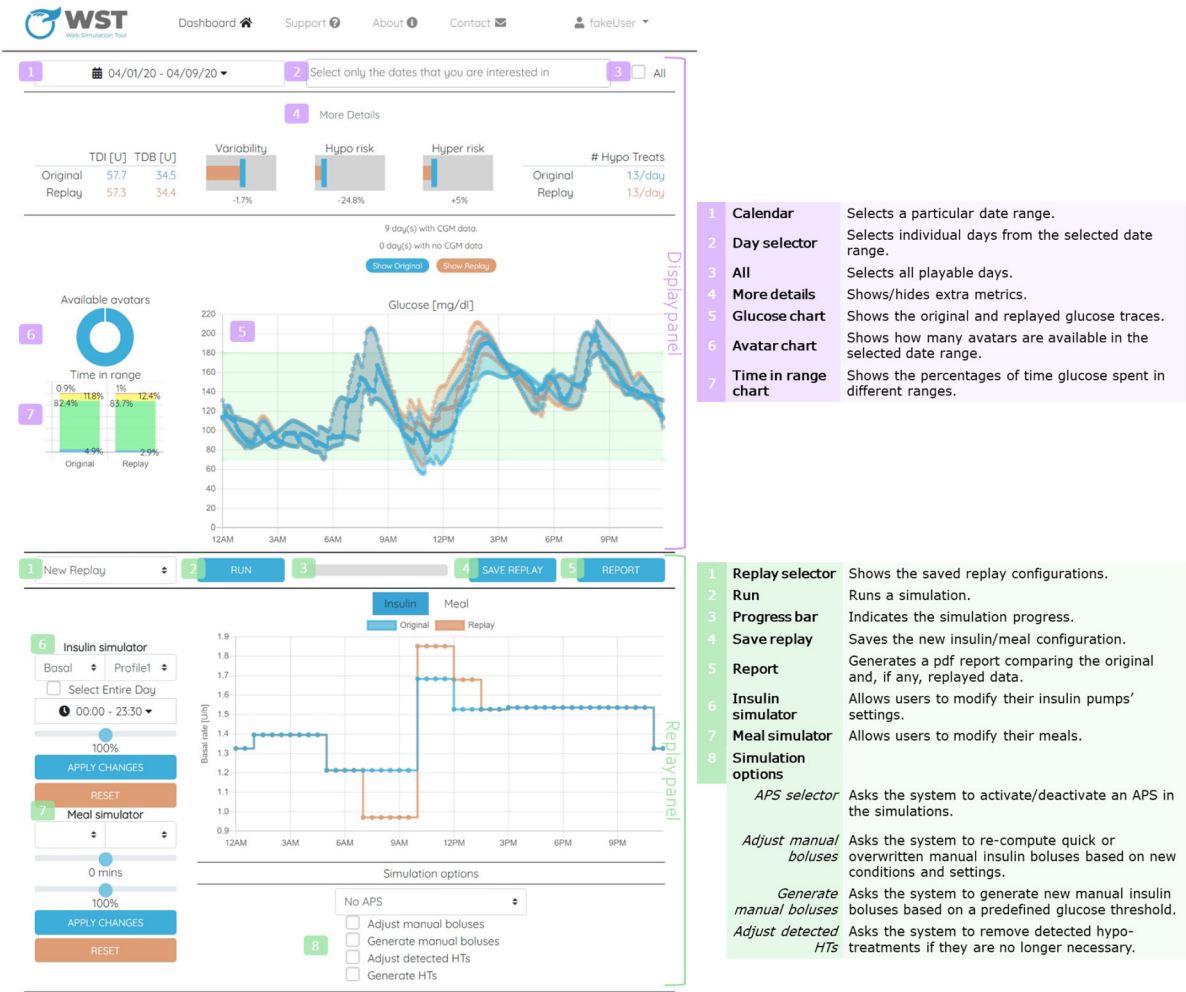


Fig. 2. Dashboard page of WST. HTs: Hypo-treatments, APS: Artificial Pancreas System.

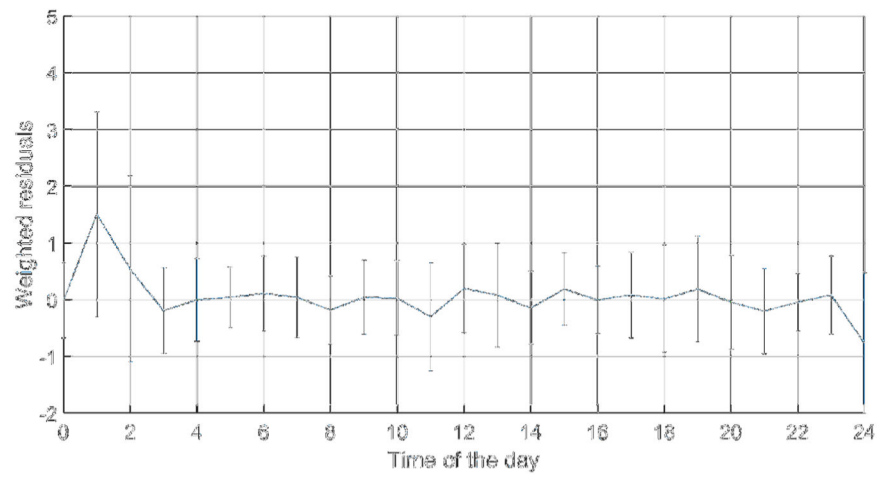


Fig. 3.
Average weighted residuals. Vertical lines indicate ± 1 SD.

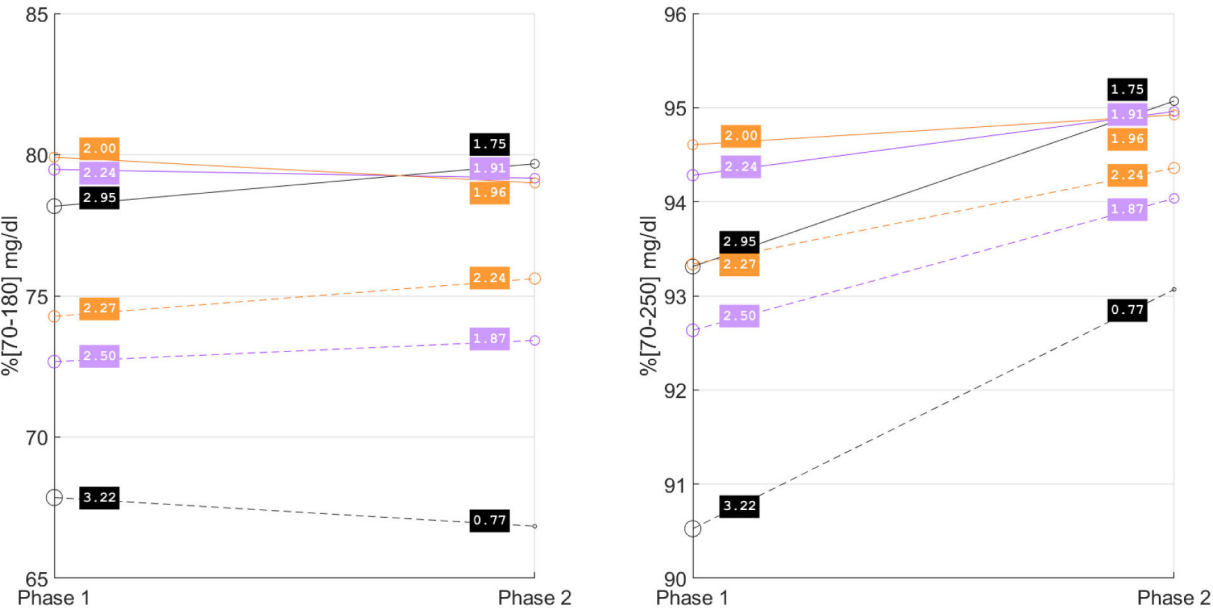


Fig. 4. Overall (violet), daytime (orange), and overnight (black) mean time in range (lines) and below 70 mg/dL (bubbles) in Phases 1 and 2 for all participants (solid) and for those who modified their pumps' settings based on WST simulations (dashed).

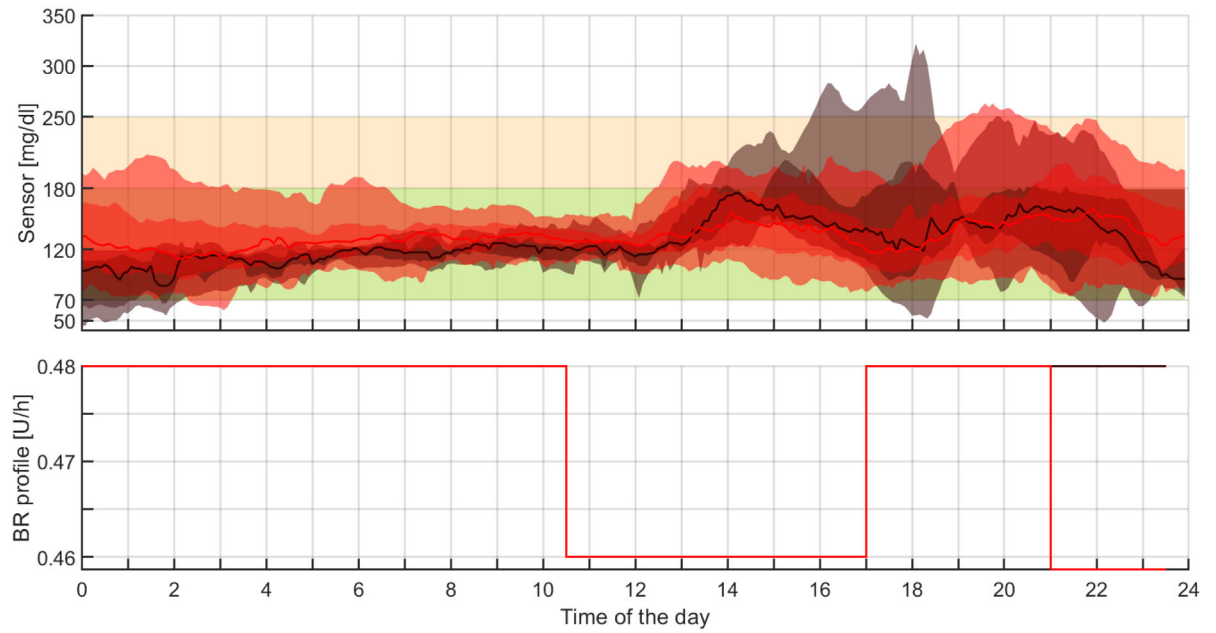


Fig. 5.

Illustrative example of therapy adaptation driven by WST simulations. Black represents Phase 1, and red, Phase 2. Above: CGM measurements. The thick lines are the median values, and the boundaries of the filled areas are the 25th and 75th percentiles (darker color), and the 5th and 95th percentiles (lighter color). Below: Mean IBR profile.



Fig. 6. Participants' responses during the follow-up interview to Q1: "*Did you enjoy using WST?*", Q2: "*Was it easy to learn how to use WST?*", and Q3: "*Do you think you have received enough training before the beginning of the trial?*".

TABLE I

EASE OF USE, TRUSTWORTHINESS, AND USEFULNESS SCORES ON A 1-5 SCALE.

Overall group	Mean	Median	IQR
Ease of use	3.2	3.0	[3.0,4.0]
Trustworthiness	3.5	4.0	[3.0,4.0]
Usefulness	3.3	4.0	[3.0,4.0]
Unaffected subgroup	Mean	Median	IQR
Ease of use	3.5	3.0	[3.0,4.0]
Trustworthiness	3.8	4.0	[3.3,4.0]
Usefulness	3.7	4.0	[3.3,4.0]

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TABLE II

COMPARISON OF MAIN GLYCEMIC METRICS BETWEEN PHASE 1 AND PHASE 2 FOR ALL PARTICIPANTS (OVERALL GROUP), AND THE SUBGROUP OF PARTICIPANTS WHO CHANGED THEIR PUMPS' SETTINGS BASED ON WST (SUBGROUP M). MEAN SENSOR GLUCOSE IS IN MG/DL AND SENSOR TIME IN %. DIFFERENCES ARE NOT STATISTICALLY SIGNIFICANT.

Overall group	Overall		Overnight	
	Phase 1	Phase 2	Phase 1	Phase 2
Mean sensor glucose	143 (20)	143 (17)	140 [121,157]	133 [120,166]
Sensor time <54 mg/dL	0.4 [0.1,0.7]	0.1 [0.0,0.7]	0.2 [0.0,0.8]	0.0 [0.0,0.0]
Sensor time <70 mg/dL	1.6 [0.7,3.7]	0.8 [0.5,3.0]	1.7 [0.0,4.2]	0.6 [0.0,2.5]
Sensor time in [70,180] mg/dL	79.5 (13.0)	79.2 (11.3)	83.2 [67.9,90.5]	85.4 [65.2,96.9]
Sensor time in [70,250] mg/dL	94.3 [91.3,95.7]	96.2 [92.1,97.9]	92.8 [88.3,98.2]	97.2 [91.8,99.6]
Sensor time >180 mg/dL	18.3 (13.5)	18.9 (11.3)	13.9 [1.2,29.0]	9.9 [2.2,33.8]
Sensor time >250 mg/dL	2.5 [0.3,7.0]	2.5 [0.2,5.4]	0.0 [0.0,8.7]	0.0 [0.0,5.6]
Subgroup M	Overall		Overnight	
	Phase 1	Phase 2	Phase 1	Phase 2
Mean sensor glucose	153 (18)	153 (13)	154 (38)	162 (30)
Sensor time <54 mg/dL	0.7 [0.5,0.9]	0.1 [0.0,0.9]	0.6 [0.0,1.7]	0.0 [0.0,0.0]
Sensor time <70 mg/dL	2.4 [1.5,3.5]	0.7 [0.3,3.4]	1.7 [0.0,6.5]	0.0 [0.0,0.9]
Sensor time in [70,180] mg/dL	72.7 (11.2)	73.4 (10.5)	67.9 (19.5)	66.8 (22.3)
Sensor time in [70,250] mg/dL	92.6 (1.8)	94.0 (4.1)	90.5 (2.7)	93.1 (6.7)
Sensor time >180 mg/dL	24.8 (12.3)	24.7 (10.1)	32.7 [9.4,48.4]	34.4 [9.0,53.4]
Sensor time >250 mg/dL	4.9 (2.9)	4.1 (2.8)	7.2 [2.9,9.6]	3.0 [0.0,12.3]